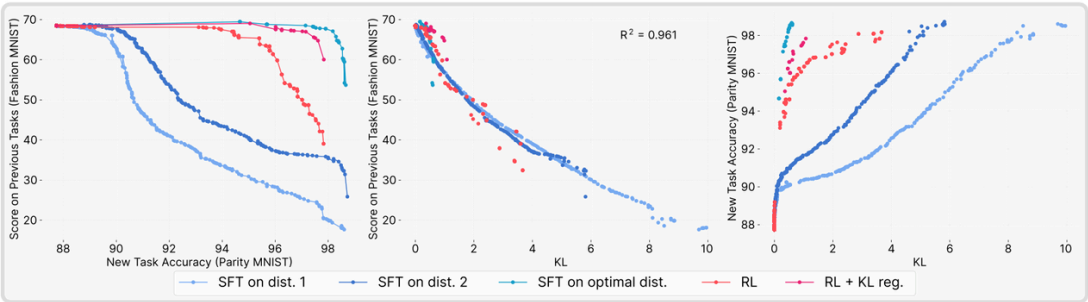
RL'S RAZOR: WHY ONLINE REINFORCEMENT LEARNING FORGETS LESS

> #RL #Theory

- Problem
  - On LLM and robotic tasks, it is often seen that even two frameworks achieve the same performance score, the model after RL maintains the original knowledge better than the model after SFT does.
- Arguments and Obervations
  - It is found that the **KL-divergence** ($\mathbb{E}_{x\sim\tau}[KL(\pi_0||\pi)]$) can be used as a measure of the severity of the forgetting.
    - ParityMNIST Experiment shows that both SFT and RL show a Forgetting-KL relationship following $R^2 = 0.96$ (mid figure below).



    - Oracle SFT Experiment (keep 100% accuracy and mathematically/manually minimize the KL between the base model and finetune model) shows that a lower KL = a better maintenance of old knowledge.
  - On-policy training (which is usually used in RL) is prone to giving a smaller KL-divergence.
    - One explanation is that on-policy takes the rollouts from the model itself, whereas the off-policy is from the labeler. Off-policy can be seen as an external distribution that is very far from the original distribution, whereas the on-policy is "from" the original distribution and try "exploring" better answers.
  - But how are on-policy methods prone to resulting in a smaller divergence?
    - One theory is that this is a sequence of **EM** algorithms.
      - Information Projection

        $Q^* = \arg\min_{Q\in\Omega} D_{KL}(Q||P)$

        Reference distribution P is fixed, and update the simple distribution Q. If complex distribution $P(\cdot)$ is very small, based on the definition of KL-divergence, $Q(\cdot) > 0$ will lead to a very large KL value (penalty). Thus, Q will be very conservative and try to fit some part of P.
      - Moment Projection

        $Q^* = \arg\min_{Q\in\Omega} D_{KL}(P||Q)$

        This is opposite to the Information Projection, where Q will try it best to fit to P to

minimize the KL divergence. Wherever there is a non-zero probability in P, Q will try to align with it.

        - A single-step in RL is an alternative occurrence of I-M projection. This will lead to, while keeping aligning with the high-accuracy distribution**s**, RL will naturally result in the best distribution with the least KL-divergence.
    - But why does SFT NOT have this alternative occurrence?
        - Because it is not **exploring FROM** the current distribution; rather, it is forced (pulled) to a distribution (label) that is very far from the current distribution, meaning I-projection is lost and only M-projection happens.
    - Alternative Theories
        - Weight-level changes: Limiting the amplitude of weight changes will reduce the forgetting. **However, based on experiment, there are weak correlations between weight change and forgetting.**
        - Representation-level changes: Some scholars argue that simply examining the weights is insufficient; instead, one should examine whether the features extracted within the model (hidden layer activation values) have changed, as maintaining these features can prevent forgetting. **However, based on the experiment, the different algorithms (SFT/RL) have different representation-drifting vs forgetting curves. This measurement is not reliable.**
        - Sparsity and rank of updates: Some researchers believe that RL is sparse during update whereas SFT is a dense update. **However, this is due to the use of `bf16`. The signal of RL is small, and sometimes the precision of `bf16` may not support these small numbers, so the system regards these numbers as 0. Changing to `fp32`, researchers found that both algorithms are dense.**
        - Distributional distances: Forward KL is usable, what about other measurements? **It is found that other measurements are usable, but not as accurate as the forward KL-divergent.**
- Work Cited
    - Paper
    - NotebookLM
    - Chat History